# Project Report

**NEXA DIGITAL SOLUTIONS**




*ICT Infrastructure & Network Consultancy*

# Abstract

The network and communication solutions discussed in this report would be implemented for South Coast Digital Services, an expanding organization that is growing from its current Wollongong office to a new Shellharbour site. The present network infrastructure is outdated since it was designed at the time when the organization was small and cannot meet today's needs, such as cloud applications, virtual meetings, file transfers, and communication platforms.

The goal of this project is to design a solid network infrastructure, which guarantees a fast communication of both locations as well as enhances network performance, security and wireless communication.

The suggested solution involves special firewall protection, managed network switches, wireless access points, and a secure connection via a site-to-site Virtual Private Network between Wollongong and Shellharbour locations.

The existing network infrastructure was reviewed to highlight possible issues such as no firewall protection, lack of wireless access, no secure office-to-office connection, and performance issues when network usage is heavy. Having evaluated the above limitations, a state-of-the-art two-offices network architecture was designed to satisfy current needs and meet possible growth in the future.

The proposed architecture will promote communication between employees, guarantee security in accessing the resources of the company, facilitate the holding of online meetings and provide a flexible infrastructure that can be adjusted in case of any future expansion of the company. In addition, the needs for reliable and manageable technologies have been considered due to the absence of an internal IT department within the organization.

In general, the project presents a viable ICT solution for South Coast Digital Services.

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# 1. Introduction

## 1.1 Project Background

South Coast Digital Services is a new provider of digital and administrative services. The company's head office is in Wollongong, and it has been steadily growing its business activities. To cater for this growth the organization will be opening a new branch office in Shellharbour to improve service delivery, increase operational capacity and serve its growing customer base.

But the company's network infrastructure was originally designed for a single-office environment and can no longer handle the growing needs of its current and future operations. This current setup does not have the flexibility, security, and management capabilities a growing multi-location organization needs.

The company is increasingly depending on Internet-based services, cloud applications, online communication platforms, email systems, and file sharing services to carry out its daily business operations. To run seamlessly, collaborate effectively and have constant access to critical business resources, these services need a stable, high-performance network environment. As the company opens a new office in Shellharbour, it needs a network solution that enables safe communication and easy data sharing between the two locations.

With the expansion, there are a few networking needs such as secure office connectivity, improved network security, traffic control, strong wireless access, and the need for future support of more users and devices. If the company doesn't improve the network infrastructure, it may run into issues such as security weaknesses, slow communication, limited sharing of resources and lower productivity.

The objective of this project is to design and implement a robust and scalable network infrastructure that connects both Wollongong and Shellharbour offices. The latest technologies in networking will be incorporated in the proposed network, which includes a robust firewall, managed switching, wireless access points, and site to site virtual private network. The proposed network infrastructure will help to facilitate secure communication between the offices, improve efficiency, promote growth, and lay down the foundation for future expansion.

### 1.2 Aim and Objectives

The main goal of the project is to establish a stable network infrastructure for South Coast Digital Services to enable the sharing of information between its two offices.

The objectives of the proposed network design include:

- Analysing the shortcomings of the existing network infrastructure of Wollongong.
- Designing a contemporary network architecture between the two offices of Wollongong and Shellharbour.
- Increasing the level of security of the network with firewall security.
- Secure communication between the offices with a site-to-site virtual private network.
- Increasing the coverage of the wireless network.
- Providing online collaboration capabilities and cloud applications.

## 2. Company Background

South Coast Digital Services is an Australian company that offers digital and administrative services to the local businesses. Currently, the company runs out of the head office situated in Wollongong. Moreover, the company is further expanding its operation by opening another branch in Shellharbour.

Currently, Wollongong office has **12 employees**, which consists of:

- 1 Director
- 1 Manager
- 1 Reception employee
- 3 Administrators
- 6 Operational employees

In addition, the new office in Shellharbour will have **8 employees**, which include:

- 1 Manager
- 3 Administrative employees
- 4 Operational employees

The organization uses computer networks, emails, cloud-based application software, file sharing and online communication for business purposes. The organization also needs reliable online meetings between the offices and external clients. And due to the expansion of two offices, there is a need for improvement in the network structure of the organization.

# 3. Current Network Analysis

## 3.1 Existing Wollongong Network Infrastructure

The existing Wollongong office network was designed when the company had fewer users and simpler requirements. The current network consists of:

- NBN business internet connection
- TP-Link broadband router
- NETGEAR 16-port unmanaged Gigabit Ethernet switch
- CAT5e Ethernet Cable
- Desktop computers
- Laptops
- Network printer
- Meeting room display
- Wireless connectivity is currently provided through the broadband router without dedicated wireless access points.

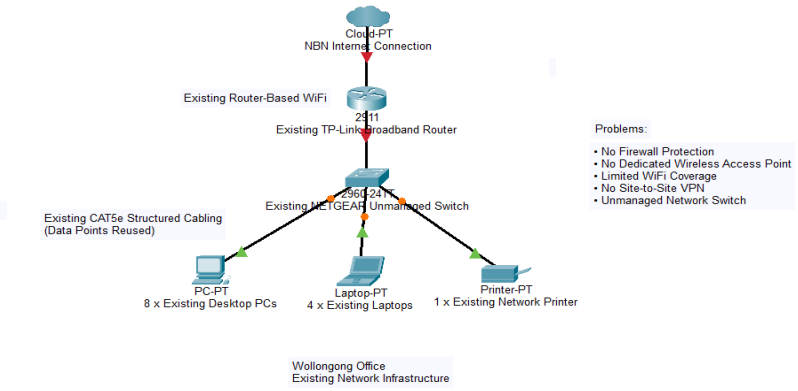



*Figure 1: Existing Network Topology*

Current network setup has a very basic design where the broadband router serves as an internet connection medium for the network. The router is connected to the unmanaged switch. The switch gives network access to computers of employees, laptops, and network printer.

There is currently a display monitor in the conference room, but no video conferencing system has been provided. There is wireless access to the internet but through the router only.

### 3.2 Current Network Limitations

The existing infrastructure has several limitations:

1. **Limited Security Protection:** The current network does not include a dedicated firewall appliance. These reduces the organization's capability to manage network traffic and secure internal resources against external threats.
2. **Limited Wireless Coverage:** Wireless connectivity is provided using the broadband router. However, this could potentially lack coverage and performance capability when the organization grows in size.
3. **No Secure Connection Between Offices:** The present network system has been designed only for one branch and hence cannot create a secure link between Wollongong and Shellharbour branches.
4. **Poor Performance During High Usage:** The client mentioned that at times of heavy loads like multiple employees working on cloud-based applications and participating in webinars at once, the internet connection performs poorly.
5. **Limited Network Management:** The unmanaged network switch can provide basic connectivity but lacks network management capabilities.

# 4. Proposed ICT Solution (Main Architecture)

### 4.1 Proposed Network Architecture Overview

This proposed ICT solution is a two-office network architecture that is secured and suitable for use by South Coast Digital Services. The proposed solution comprises:

- Internet access at a business level

- provision of firewall protection in both offices
- Network switches managed
- Wireless access points
- Site to site virtual private network connection for Wollongong and Shellharbour proves communication ability in meeting rooms.

This architecture allows the independent operations of the offices with safe communication and resource sharing between the two.

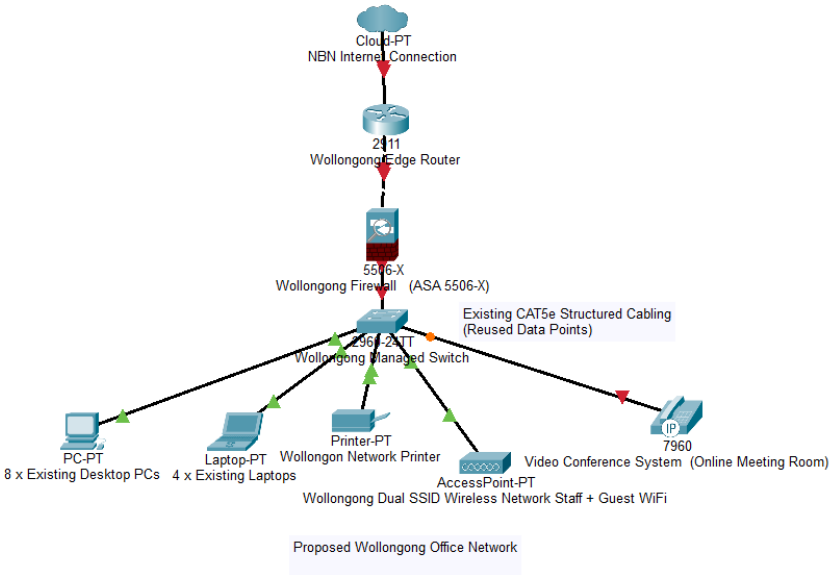


*Figure 2: Proposed Wollongong Office Network Architecture*

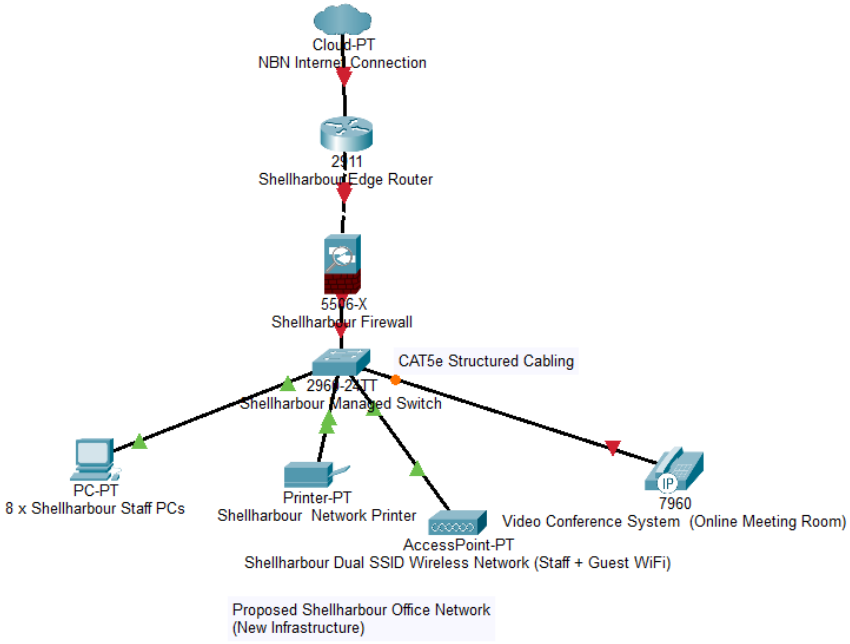



*Figure 3: Proposed Shellharbour Office Network Architecture*

The network connects the Wollongong and Shellharbour offices using the site-to-site virtual private network connection.

The Wollongong office uses:

- Edge router
- Firewall
- Managed switch
- Wireless access point
- Existing computers and laptops
- Network printer
- Video conferencing system
- The Shellharbour office uses:
- Edge router
- Firewall

- Managed switch
- Wireless access point
- Computers for staff
- Network printer
- Communication system in meeting room

The virtual private network connection provides safe communication between both offices whereby employees can access authorized company resources.

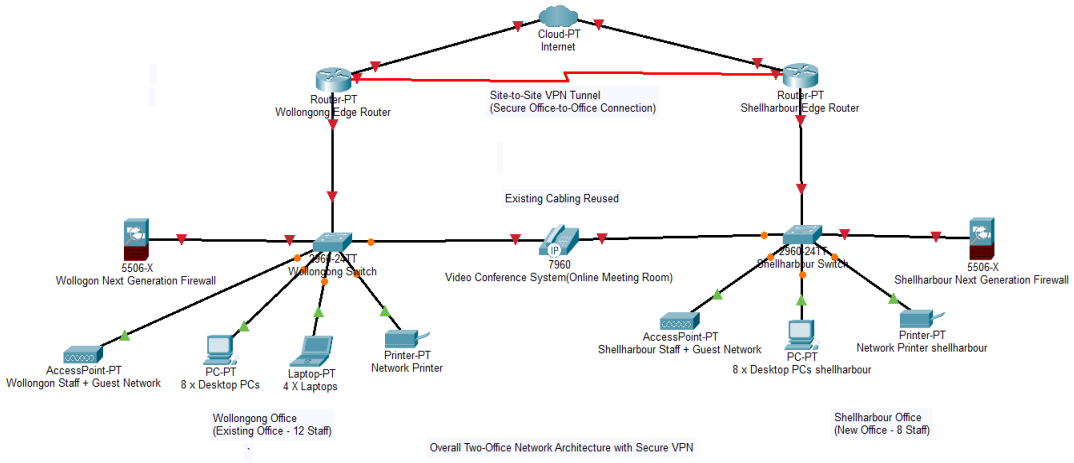



*Figure 3: Overall, Two-Office Architecture*

This architecture gives a clear picture of the solution for the network.

This network solution facilitates effective communication between the two premises by ensuring that the network is integrated. The employees from Wollongong and Shellharbour will be able to communicate in a secure manner using the VPN connection, while still accessing cloud applications, file transfer sites, and virtual meeting rooms.

Moreover, the proposed network architecture can facilitate growth in the future since more devices and services can be added without changing the network completely.

# 5. Hardware Recommendations

## 5.1 Firewall

It is recommended to use dedicated firewall equipment in both offices.

The benefits include:

- Traffic filtering on the network
- Defence from external dangers
- Secure establishment of VPN connection
- Better management of access to the network

## 5.2 Managed Switches

It is preferable to have managed switches rather than unmanaged switches.

The benefits include:

- Improved network management
- Increased capability of monitoring
- Troubleshooting
- Expansion

## 5.3 Wireless Access Points

It is advisable to install wireless access points in both offices.

The benefits include:

- Wireless coverage improvement
- Improved performance of staff devices
- Separate wireless networks for staff and visitors
- Improved network accessibility of visitors

### 5.4 Site-to-Site VPN

It is advisable to set up a site-to-site VPN connection between Wollongong and Shellharbour.

The benefits include:

- Data communication security
- Security of business data
- Collaboration between offices
- Multilocation collaboration for employees

## 6. Network Technology Justification

The selected technologies are those which are effective in addressing the current limitations of the company's existing network infrastructure and which support the future growth and operational requirements of South Coast Digital Services.

Implementation of the next generation firewall will significantly improve the level of security for the network as opposed to the present-day simple router security setup. The new firewalls, as opposed to the old routers, come equipped with security features that include intrusion detection, monitoring of traffic, application control, anti-malware and access control policies. All these will ensure the protection of corporate data from any cyber-attacks that may arise due to reliance on digital services.

Replacement of the current unmanaged switches by managed switches will enable increased flexibility and efficiency in network management. It will be possible to use the managed switches to monitor traffic in the network, set VLANs, give priority to applications via Quality of Service and identify any network problems. This will ensure that the company continues experiencing improved network performance in future.

Deployment of wireless access points will provide reliable and wider wireless coverage across the office locations. This will enhance connectivity for employees with laptops, mobile devices and other wireless-enabled equipment while providing controlled guest access for visitors. With

the right wireless configuration and security, the company can deliver the convenience of connectivity and not sacrifice network protection.

As the number of offices in South Coast Digital Services has increased from one office to several others, there has been a need for a secure means of communicating amongst the offices. Site to site VPN technology is going to be utilized to establish a secure connection among the branch offices allowing the company employees at different places to exchange information and communicate via the secure network. Thus, there will be less dependence on the internet and business information will remain confidential during the process of its transmission.

The suggested network topology is also scalable and forward-looking. There is an opportunity for the expansion of the user base, devices, offices and network services without the total reconfiguration of the network. The suggested network topology incorporates advanced security measures, managed network devices, reliable wireless connectivity and secure inter-office communications.

The network infrastructure proposed would therefore provide South Coast Digital Services with a robust, secure, manageable and scalable system that can support the current business operations and prepare the company for future technology growth and increased connectivity demands.

# 7. Requirement Analysis

## 7.1 Business Requirements

South Coast Digital Services requires a state-of-the-art network infrastructure that is able to support the growth of the company by expanding itself from one-office to two-offices organization. At present, South Coast Digital Services relies heavily on cloud applications, emails, online meetings, and file sharing capabilities for performing the work process. Hence, the new network must ensure seamless connectivity and secure communication as well as effective resource access between Wollongong and Shellharbour offices.

- The key business requirements include:
- Secure communication between two offices
- Reliable access to cloud applications

- Online meeting and collaboration facility
- Secure file sharing among employees
- Better wireless connection facility
- Remote access facility for off-site employees

The future-proof network should be scalable enough to accommodate extra users and devices and services in future without replacing the entire network.

## 7.2 Communication Requirements

Currently, email, mobile phone, online messaging, and sometimes online meeting is used for communication. The introduction of Shellharbour office has increased the need for communication between both offices.

The proposed solution should support:

- This new technology must be able to facilitate:
- Video coding for staff meetings
- Client communication
- Instant messaging and collaboration
- Secure access to the resource

Connection through Site-to-Site VPN link between both offices can enable secure communication and access to authorized company resources.

## 7.3 Existing Equipment Evaluation and Upgrade Requirements

The current network equipment installed in the Wollongong office is fit for a small environment. With the growth of the company into a two-office network, current equipment will be reviewed in order to decide whether it will still be able to cater for the organization's needs. The current desktops, laptops, and network printers will be kept wherever possible since they can still be used in performing daily operations including use of cloud applications, file transfer, and online communication. However, the current unmanaged switch and simple broadband router will be upgraded to managed switches and firewall equipment respectively. This will make the network more secure, monitorable, manageable, and scalable in future. This strategy will allow South

Coast Digital Services to avoid unnecessary expenses in equipment replacement while upgrading necessary network equipment.

# 8. Office Implementation Design & Security

This section presents the practical deployment design for the Wollongong and Shellharbour offices. The focus is on physical network implementation, placement of networking equipment, wireless connectivity, printer integration, meeting room connectivity, and security considerations required for successful deployment.

## 8.1 Wollongong Office Network Layout

This is the current office layout for the main office of South Coast Digital Services. The suggested design enhances the current setup by having a special network cabinet that contains the router, firewall, and managed switch. The existing cable and network ports may be utilized where possible to minimize costs and installation disturbance. Wireless access points have been located to cover the whole office.

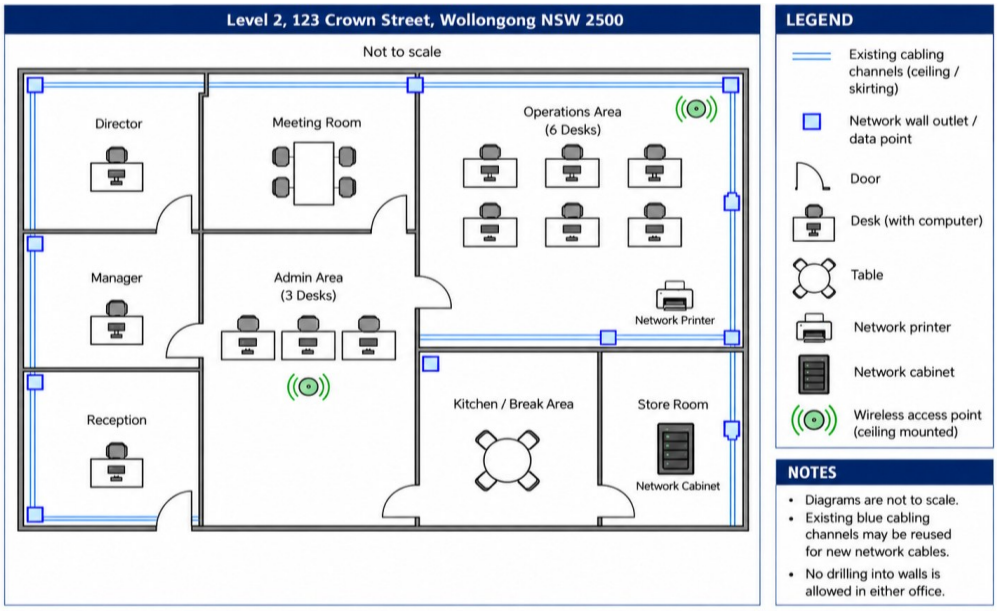



*Figure 4: Proposed Wollongong Office Network Layout*

The floor plan provides network connectivity for employee workstations, meeting room facilities, printer services, and wireless devices. The network cabinet acts as the central point for managing wired and wireless communication.

## 8.2 Shellharbour Office Network Layout

The Shellharbour branch is an entirely new branch that requires a whole network setup. The suggested design consists of the following components: router, firewall, managed switch, Wi-Fi, workstations for employees, printer connection, and communication in the conference room. The configuration will be able to cover 8 employees.

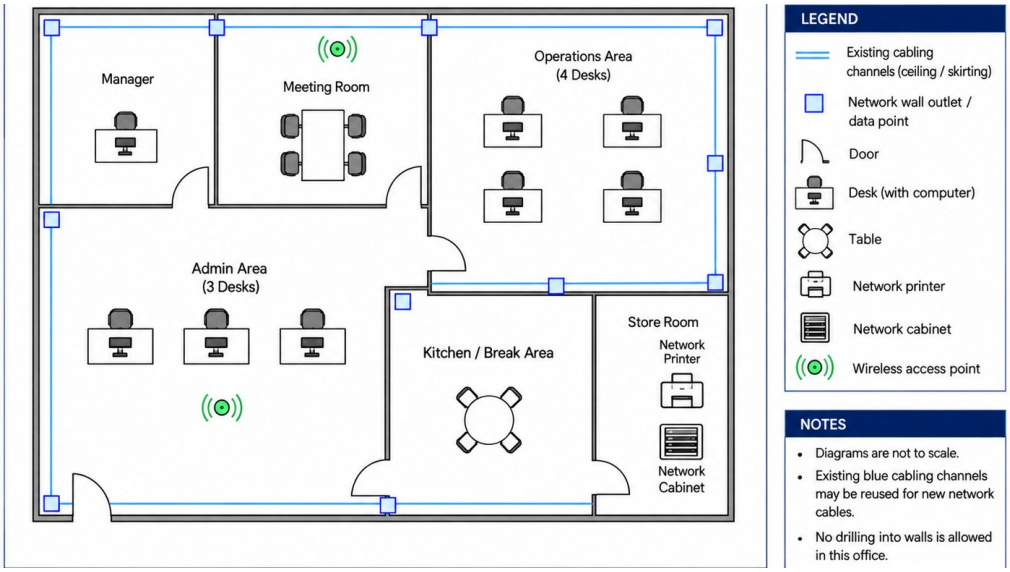



*Figure 5: Proposed Shellharbour Office Network Layout*

The network rack is in a secure area and contains the main networking equipment. The managed switch connects computers, printers, and meeting room devices via CAT6 cabling. Wireless access points enable flexible connectivity.

## 8.3 Practical Deployment of the Network Solution

The deployment process starts with the preparation of the network sites and equipment requirements check. Hardware installation comprises of routers, firewalls, managed switches,

Wi-Fi Access Points, printer connection, and conference room connectivity. Once the hardware is installed, the network devices are configured and tested for reliable functioning.

Security configuration covers firewall policies, secured communication via a VPN from one office to another, configuration of wireless network, isolation of employees' and guests' WiFi, and access to internal business assets.

### 8.4 Security Implementation Overview

Security is incorporated into the design of the network through firewall, site-to-site secure connection, traffic segregation using VLANs, and wireless security. Firewall is designed to protect internal systems from accessing network traffic and unauthorized access.

The use of separate wireless networks for staff and visitors helps secure the system since it prevents guests from accessing company's internal resources. This will be a suitable security solution for South Coast Digital Services since it guarantees them safe and manageable network.

The use of remote access via a virtual private network enables the firm to provide safe access to its internal resources by remote employees.

## 9. Implementation Plan

The implementation of the proposed network solution will be completed in several stages to minimise disruption to business operations.

### 9.1 Stage 1: Network Preparation

Stage one is to prepare both locations before installation. It will include an evaluation of the current infrastructure at Wollongong, as well as preparing new equipment needs at Shellharbour.

Activities include:

- Identifying network equipment needs
- Evaluating existing cabling/data points
- Specifying internet service providers

- Preliminary network installation sites preparation

## 9.2 Stage 2: Hardware Installation

The next step is the installation of the suggested networking equipment.

- Installation entails:
- Installation of firewall
- Installation of managed switches
- Installation of wireless access points
- Router installation
- Networking of printer
- Setup of meeting room connectivity

Complete installation of networking equipment is needed for Shellharbour office since there is no networking equipment installed.

## 9.3 Stage 3: Configuration and Security Setup

Following installation of the hardware, network equipment will then be configured, and configurations will be done as follows:

- Firewall security policies
- Connection using a Virtual Private Network (VPN) between office branches
- Wireless network configuration
- Staff and guests Wi-Fi networks segregation
- Network access control

The above configurations will enhance security to ensure that only authorised personnel can access the company's facilities.

## 9.4 Stage 4: Testing and Staff Training

The network will be tested before being put into use for reliability.

This will include:

- Internet connectivity testing
- VPN communications testing
- Wireless range testing
- Online meetings testing
- File access testing

In addition, staff training will be conducted to make sure that the employees know how to use the network properly.

# 10. Cost Analysis

The estimated cost for the network upgrade has been done in a manner that would make sure that it stays within the upper budget limit of $60,000 by the firm.

## 10.1 Estimated Cost Breakdown

| Item | Estimated Cost |
|---|---|
| **Firewalls (two offices)** | $6,000 |
| **Managed switches** | $4,000 |
| **Wireless access points** | $2,500 |
| **Routers and network equipment** | $3,000 |
| **VPN configuration** | $1,500 |
| **Installation and configuration** | $8,000 |
| **Internet services (24 months)** | $15,000 |
| **Testing and staff training** | $2,000 |
| **Additional cables/accessories** | $2,000 |
| Estimated Total | **$44,000** |

The proposed solution remains below the allocated budget while providing additional capacity for future expansion.

### 10.2 Installation Cost Calculation

The installation cost is calculated according to the required technician rate of $160 per hour.

Estimated technician hours:

| Task | Hours |
|---|---|
| **Hardware installation** | 20 hours |
| **Firewall and VPN configuration** | 15 hours |
| **Switch and wireless configuration** | 10 hours |
| **Testing** | 5 hours |
| **Staff training** | 5 hours |
| **Total** | 55 hours |

Total labour cost:

$$55 \times \$160 = \textbf{\$8,800}$$

# 11. Conclusion

The suggested ICT solution will ensure a safe, reliable, and scalable network infrastructure that is able to successfully operate within the context of the expansion of the organization to several office locations. Implementation of the advanced network solutions such as dedicated firewalls, managed switches, wireless access points, and site-to-site Virtual Private Network (VPN) will solve the most serious problems of the current network environment.

Dedicated firewall will provide the required protection of the network, will control traffic flows, will protect company's assets from any outside threats, and will allow managing access to the system. Managed switches will help to manage network better because of more sophisticated configuration and scalability in the future.

Moreover, the establishment of a secure connection via the use of a secure VPN connection between the Wollongong and Shellharbour office facilities will allow the staff to effectively communicate and use authorized company resources. Therefore, various business tasks, including cloud-based applications, remote online meetings, file exchange, and collaboration

between offices will become easier. Thus, it is expected that the employees will have higher productivity owing to faster and more reliable access to the necessary resources.

The suggested network design is flexible in terms of further development. The network may be scaled, and additional users, equipment, and network services may be implemented to it without changing the existing structure of the network.

In summary, the suggested network upgrade solution gives South Coast Digital Services the ability to gain a solid technological base in terms of security, communication, performance, and operation management. This solution will help to solve the existing problems related to the current network and prepare the company for future development.

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